Project Support Session

Personal Loan Campaign

Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Submission Guidelines
- FAQs

Personal Loan Campaign - Business Context

AllLife Bank is a US bank that has a growing customer base. The majority of these customers are liability customers (depositors) with varying sizes of deposits. The number of customers who are also borrowers (asset customers) is quite small, and the bank is interested in expanding this base rapidly to bring in more loan business and in the process, earn more through the interest on loans. In particular, the management wants to explore ways of converting its liability customers to personal loan customers (while retaining them as depositors).

A campaign that the bank ran last year for liability customers showed a healthy conversion rate of over 9% success. This has encouraged the retail marketing department to devise campaigns with better target marketing to increase the success ratio.

You as a Data Scientist at AllLife Bank have to build a model that will help the marketing department to identify the potential customers who have a higher probability of purchasing the loan.

Personal Loan Campaign - Objective

As a Data Scientist at AllLife Bank, your task is to develop a Machine Learning model that addresses the bank's marketing challenge. The objective is to predict whether a liability customer will purchase a personal loan, understand which customer attributes are most significant in driving purchase decisions, and identify which segment of customers to target to rapidly expand the loan business.

Personal Loan Campaign - Data Dictionary

Name	Description
ID	Customer ID
Age	Customer’s age in completed years
Experience	# years of professional experience
Income	Annual income of the customer (in thousand dollars)
ZIPCode	Home Address ZIP code.
Family	The family size of the customer
CCAvg	Average spending on credit cards per month (in thousand dollars)
Education	Education Level. 1: Undergrad; 2: Graduate; 3: Advanced/Professional

Personal Loan Campaign - Data Dictionary

Name	Description
Mortgage	Value of house mortgage if any. (in thousand dollars)
Personal_Loan	Did this customer accept the personal loan offered in the last campaign?
Securities_Account	Does the customer have a securities account with the bank?
CD_Account	Does the customer have a certificate of deposit (CD) account with the bank?
Online	Do customers use Internet banking facilities?
CreditCard	Does the customer use a credit card issued by any other Bank (excluding All Life Bank)?
Mortgage	Value of house mortgage if any. (in thousand dollars)
Personal_Loan	Did this customer accept the personal loan offered in the last campaign?

Personal Loan Campaign - Q&A

Personal Loan Campaign - Evaluation Rubric

Section	Points
Define the Problem & Exploratory Data Analysis	10
Data Preprocessing	6
Decision Tree Model Building	10
Model Performance Evaluation & Improvement	20
Actionable Insights & Recommendations	6
Presentation/Notebook – Overall Quality	8
Total	60

Personal Loan Campaign - Evaluation Rubric

Section 1: Define the Problem & Perform EDA

What to Cover?	Guiding Questions and Considerations
Problem definition, data background	Clearly define the business problem & objectives. Check data sanity.
Univariate analysis & Bivariate analysis	What patterns emerge from age, income, credit usage, mortgage, family size, education, etc.?
Perform complete univariate & bivariate EDA with meaningful observations.	Which customer groups display a higher purchase probability? Are certain behaviors strongly correlated with Personal_Loan?

Personal Loan Campaign - Evaluation Rubric

Section 2: Data Pre-processing

What to Cover?	Guiding Questions and Considerations
Handle missing values & outliers (if any).	Are there missing values or outliers that need correction, imputation, or removal?
Feature Engineering	Do features require encoding, scaling, or transformation?
Data preparation for modelling	Have we separated the target variable from the input data? Have we split the data into training and testing sets?

Personal Loan Campaign - Evaluation Rubric

Section 3: Decision Tree Model Building

What to Cover?	Guiding Questions and Considerations
Define model evaluation criterion	Which evaluation metric is most appropriate?
Build the model and comment on the model performance	Is the baseline model overfitting or underfitting, and how is the model performing?
Visualize the decision rules and important features	Does the tree visualization reveal any dominant splits, thresholds, or patterns that are useful for business interpretation?

Personal Loan Campaign - Evaluation Rubric

Section 4: Model Performance Evaluation & Improvement

What to Cover?	Guiding Questions and Considerations
Try and improve the model performance by pruning (Both Post and Pre pruning)	• Does pruning improve generalization and reduce overfitting? ○ If yes, then which model configuration performs the best?
Check the performance of the pruned models, compare the performance of all the models built, and select the final model	How do model performance metrics compare with each other?
Find the decision rules and check feature importance for the final model	Which feature is the most important? Verify that the feature importance values are consistent with the decision tree visualization.

Personal Loan Campaign - Evaluation Rubric

Section 5: Actionable Insights & Recommendations

What to Cover?	Guiding Questions and Considerations
Conclude with the key takeaways for the marketing team	Which customer segments show the highest likelihood to take the loan?
What would your advice be on how to do this campaign?	What targeting strategies and customer engagement approaches can maximize campaign ROI?

Personal Loan Campaign - Evaluation Rubric

Section 6: Presentation/Notebook - Overall Quality

Low-code Considerations	Full-code Considerations
Clear structure, flow, and visual appeal - everything sits well in a story	Clear structure and flow
Crispness - Focus on key points	Well-commented and executed code
All sections of the rubric included, even if in an appendix	All sections of the rubric included, even if in an appendix

Personal Loan Campaign - Low-Code Version

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Learner Notebook - Low Code* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

Personal Loan Campaign - Low-Code Version

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

Personal Loan Campaign - Full-Code Version

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution codes from scratch

Download the dataset and the *Learner Notebook - Full Code* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Clearly write down observations, insights, and recommendations for the business problem based on the analysis performed

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

Personal Loan Campaign - Full-Code Version

The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)

The conversion can be done via one of the following ways:

Jupyter Notebook: ***Download as HTML*** from the ***File*** menu
Google Colab: Use free online tools

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

Personal Loan Campaign - FAQ

Why are arrows not appearing in my Decision Tree visualization, and how do I enable them?

Use the following code as a reference to resolve the issue and make necessary changes (name of the model, feature names, etc):

```
plt.figure(figsize=(20,30))
Out = tree.plot_tree(model,feature_names=feature_names,
filled=True,fontsize=9,node_ids=False,class_names=None,)
#below code will add arrows to the decision tree split if they are missing
for o in out:
     arrow = o.arrow_patch
     if arrow is not None:
        arrow.set_edgecolor('black')
        arrow.set_linewidth(1)
 plt.show()
```

Personal Loan Campaign - FAQ

How to deal with "ZIPCode" as it is a numeric value but it's also essentially a category?

You can explore the following links to deal with zip codes:

1. uszipcode - Python package that can help in mapping zip codes to different locations

2. https://www.smartystreets.com/articles/zip-4-code - Description of how zip codes are created in the US.

Personal Loan Campaign - FAQ

I'm trying to post-prune the decision tree. But I'm getting the following error:

```
"ValueError: ccp_alpha must be greater than or equal to 0"
```

To resolve this error kindly use absolute values (positive value) of alpha. Use the following lines of code to resolve the error:

```
ccp_alphas, impurities = abs(path.ccp_alphas), path.impurities
```

Power Ahead!